

Chapter 18 - The AHX Engine

AHX (Abyss High-Quality Music) is the four-channel chip-tune format from the Amiga scene of the mid-1990s. It uses a small synthesis core based on the Amiga Paula chip to produce its distinctive square-and-saw sound without storing any sample data. Intuition Engine's AHX engine plays AHX files directly: load the data, set the play pointer, and write to the control register.

18.1 What AHX can show

Item	Value
Voices	4, each driven by the AHX synthesis engine
File format	Standard AHX (. ahx) modules
Subsongs	Up to 256 per file
Output	Routed into the global mixer

18.2 The register block

The engine has a tiny register footprint at 0xF0B80–0xF0B91.

Address	Name	Purpose
0xF0B80	AHX_PLUS_CTRL	AHX+ mode enable (bit 0).
0xF0B84	AHX_PLAY_PTR	Address of the AHX data in memory.
0xF0B88	AHX_PLAY_LEN	Length of the AHX data in bytes.
0xF0B8C	AHX_PLAY_CTRL	Bit 0 = start, bit 1 = stop, bit 2 = loop.
0xF0B90	AHX_PLAY_STATUS	Bit 0 = busy, bit 1 = error.
0xF0B91	AHX_SUBSONG	Subsong number (0–255).

18.2.1 The AHX+ extension

Bit 0 of AHX_PLUS_CTRL enables **AHX+**: an IE-specific extension that adds additional waveforms and an effects layer on top of the standard AHX synthesis core. Bit-exact AHX behaviour resumes when the bit is cleared.

18.3 BASIC keywords

Form	Effect
AHX PLAY $addr[, len[, subsong]]$	Start AHX playback from $addr$.
AHX STOP	Stop playback.
AHX PLUS ON / AHX PLUS OFF	Enable / disable AHX+.

A BASIC fragment that plays an AHX module already loaded into memory at &H100000:

```
10 POKE &H000F0800, 1          : REM AUDIO_CTRL = on
20 AHX PLAY &H100000, &H4000    : REM 16 KB module
```

To play a file from disk in one step, use `SOUND PLAY` with the `.ahx` extension (Chapter 22).

18.4 Putting it together

AHX is a small, fast engine. It costs almost no CPU time to keep running once started; the engine renders each frame internally and feeds the mixer. Two practical uses: background music for games (start once, let it loop), and effect-driven music demos (change `AHX_SUBSONG` to switch between intro and main themes without reloading).

The next chapter covers ProTracker MOD playback, the four-channel sample-based Amiga format that complements AHX.